Practical No. 3

Services, Notifications and Broadcast Receivers

1. Develop an Android application implementing Started and Bound Services.
2. Develop an Android application implementing Notifications using Notification Manager.
3. Develop an Android application implementing Static and Dynamic Broadcast Receivers.

AIM: To study and implement Programs on Services, notification and broadcast receivers Programs on Services, notification and broadcast receivers

- Started Service
- Notification from Service
- Broadcast Receiver (BOOT_COMPLETED)
- Buttons to Start/Stop Service
- Notification Channel (Android 8+)

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest package="com.example.servicedemo"
    xmlns:android="http://schemas.android.com/apk/res/android">
    <!-- Permission for Boot Completed -->
    <uses-permission
android:name="android.permission.RECEIVE_BOOT_COMPLETED"/>
    <application
        android:allowBackup="true"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">
        <!-- Service -->
        <service android:name=".MyService"/>
        <!-- Broadcast Receiver -->
        <receiver android:name=".BootReceiver"
            android:enabled="true"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.BOOT_COMPLETED"/>
            </intent-filter>
```

```
        </receiver>
        <!-- Main Activity -->
        <activity android:name=".MainActivity">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>
    </application>
</manifest>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <Button
        android:id="@+id/btnStart"
        android:text="Start Service"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/btnStop"
        android:text="Stop Service"
        android:layout_marginTop="10dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
```

```
    <Button
        android:id="@+id/btnNotify"
        android:text="Send Notification"
        android:layout_marginTop="10dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.servicedemo;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
    Button btnStart, btnStop, btnNotify;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnStart = findViewById(R.id.btnStart);
        btnStop = findViewById(R.id.btnStop);
        btnNotify = findViewById(R.id.btnNotify);
        // Start Service
        btnStart.setOnClickListener(v -> {
            Intent intent = new Intent(this, MyService.class);
            startService(intent);
            Toast.makeText(this, "Service Started", Toast.LENGTH_SHORT).show();
        })
```

```
        // Stop Service
        btnStop.setOnClickListener(v -> {
            Intent intent = new Intent(this, MyService.class);
            stopService(intent);
            Toast.makeText(this, "Service Stopped", Toast.LENGTH_SHORT).show();
        });
        // Send Notification via Service
        btnNotify.setOnClickListener(v -> {
            Intent intent = new Intent(this, MyService.class);
            intent.setAction("SEND_NOTIFICATION");
            startService(intent);
        });
    }
}
```

MyService.java

```
package com.example.servicedemo;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.app.Service;
import android.content.Intent;
import android.os.Build;
import android.os.IBinder;
import androidx.core.app.NotificationCompat;
public class MyService extends Service {
   private static final String CHANNEL_ID = "ServiceChannel";
  @Override
    public int onStartCommand(Intent intent, int flags, int startId) {
if (intent != null && "SEND_NOTIFICATION".equals(intent.getAction())) {
            sendNotification()  }
```

```
 return START_STICKY;
 }
private void sendNotification() {
 NotificationManager manager =
   (NotificationManager) getSystemService(NOTIFICATION_SERVICE);
// Notification Channel (For Android 8+)
     if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
        NotificationChannel channel = new NotificationChannel(
              CHANNEL_ID,
              "Service Channel",
              NotificationManager.IMPORTANCE_DEFAULT);
        manager.createNotificationChannel(channel);
     }
  NotificationCompat.Builder builder =
           new NotificationCompat.Builder(this, CHANNEL_ID)
                .setContentTitle("Service Notification")
                .setContentText("This is a Notification from Service")
                .setSmallIcon(android.R.drawable.ic_dialog_info)
                .setAutoCancel(true);
 manager.notify(1, builder.build());
  }
@Override
  public IBinder onBind(Intent intent) {
     return null;
  }}
```

BootReceiver.java

```
package com.example.servicedemo;
import android.content.BroadcastReceiver;
import android.content.Context;
```

```
import android.content.Intent;
import android.widget.Toast;
public class BootReceiver extends BroadcastReceiver {
   @Override
    public void onReceive(Context context, Intent intent) {
if (Intent.ACTION_BOOT_COMPLETED.equals(intent.getAction())) {
     Toast.makeText(context,
           "Device Booted - Broadcast Received",
               Toast.LENGTH_LONG).show();
 // Optional: Start Service Automatically
         Intent serviceIntent = new Intent(context, MyService.class);
         context.startService(serviceIntent);
      }
   }}
```



